

BCI기반 뇌파 제어 자율주행차 시스템

Motor Imagery-Based EEG Autonomous Vehicle Control System

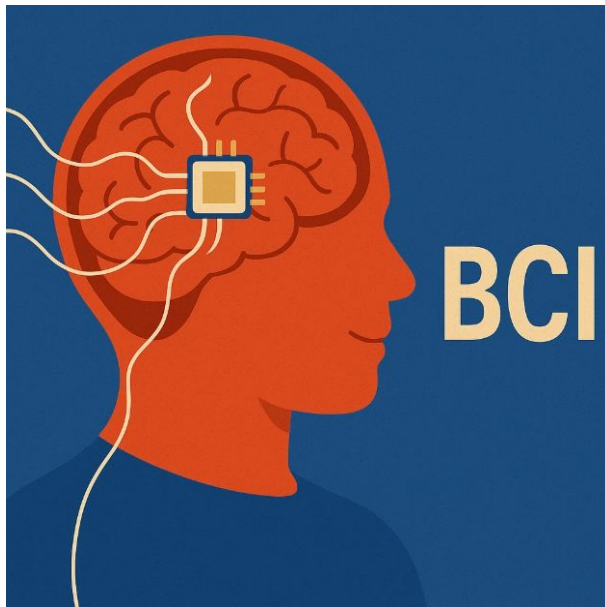
23011869 강민우, 26012205 노승민, 26010270 주수연

연구 배경



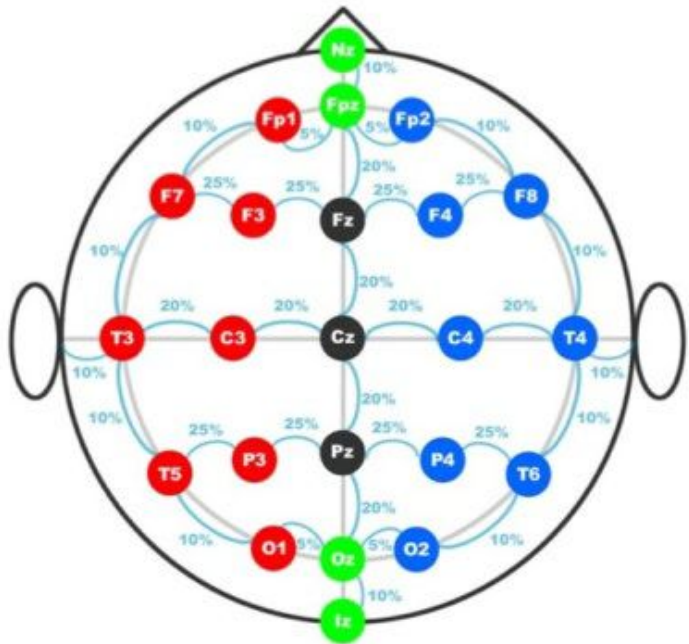
연구 동기

왜 뇌파?



연구 목표

10 / 20 System Electrode Distances

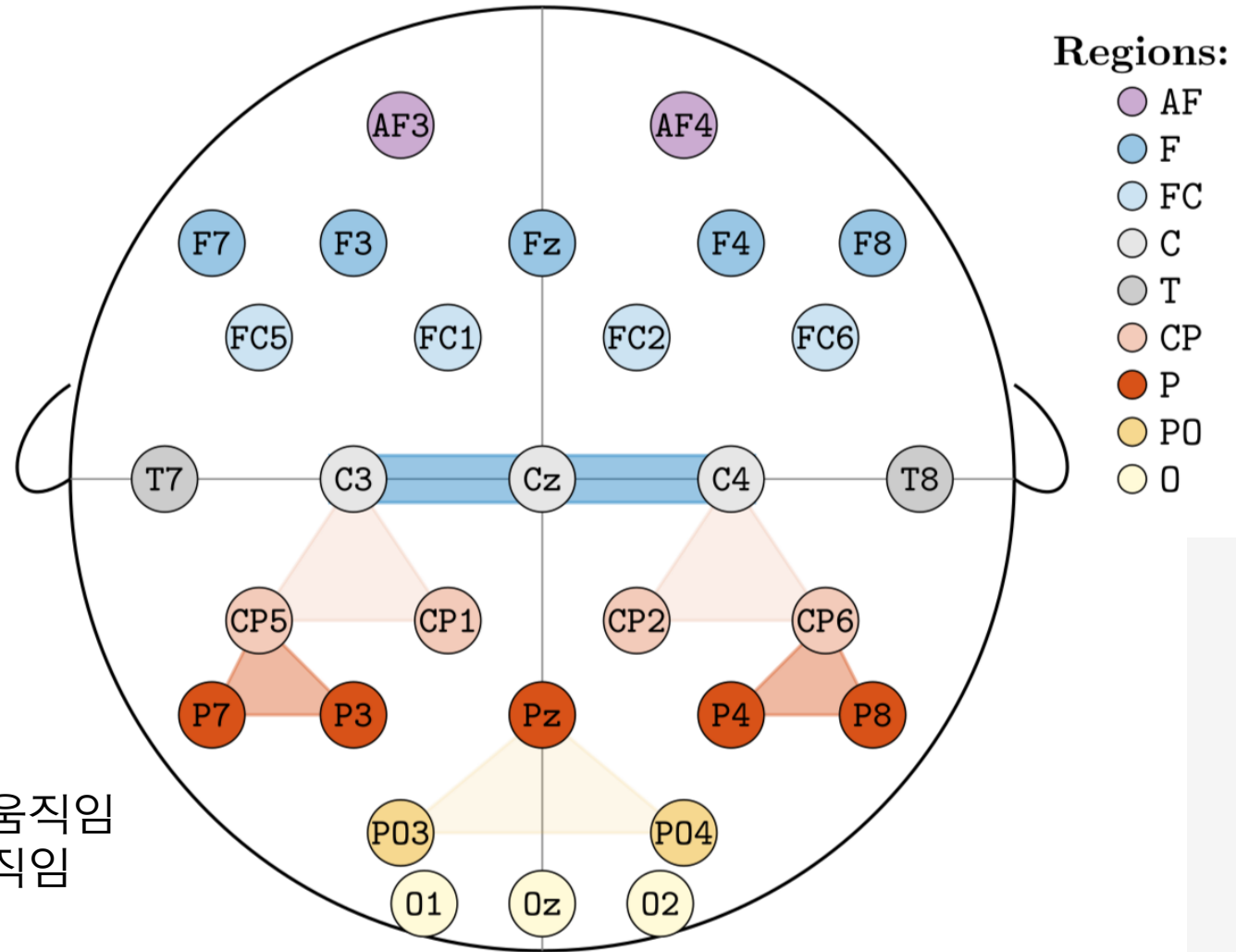


좌회전, 우회전, 정지

BCI

델타파 (0.5~4 HZ) : 깊은 수면
세타파 (4~8 HZ) : 얇은 수면, 이완
알파파 (8~13 HZ) : 안정, 무 리듬
베타파 (13~30 HZ) : 활동, 운동 계획
감마파 (30+ HZ) : 고차원 인식

노파

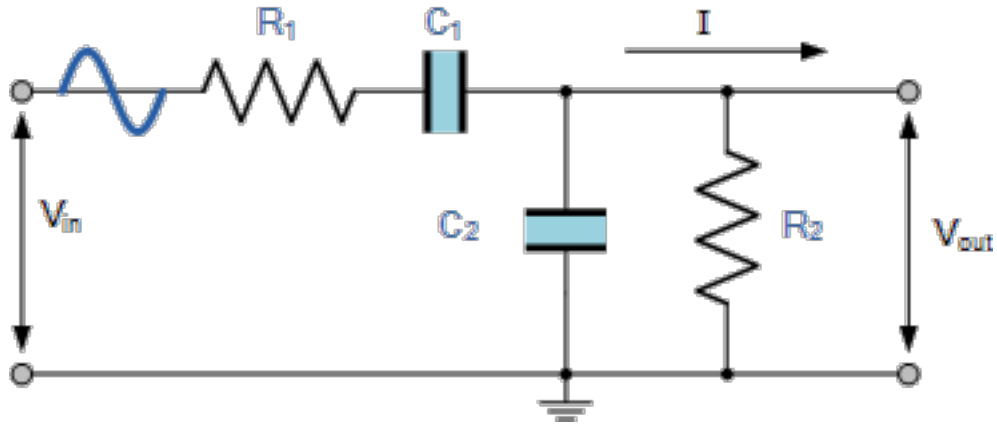


C3 : 오른쪽부분의 움직임
C4 : 왼쪽부분의 움직임

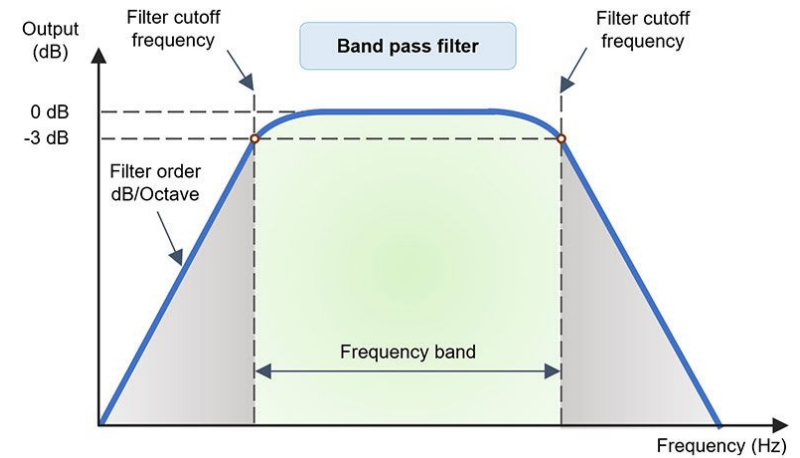
노이즈



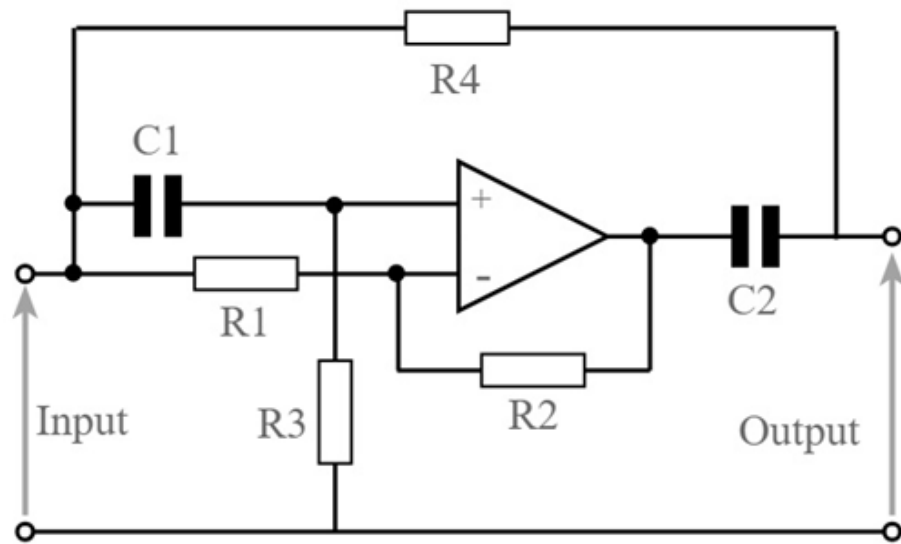
Band Pass Filter



대역의 주파수를 감쇄시키는 필터



노치 필터



특정 주파수 대역의 신호를 감쇠하고
이 대역 위 및 아래의 신호를 통과

독립 성분 분석(EEG)

$$y(n) = \sum_{i=1}^N f(n) \quad \text{뇌파 신호(시계열 형태)}$$



$$x(n) = \sum_{i=0}^N f(n) * w(n)$$

윈도우 함수 $w(n)$ 을 곱하여 불연속적인 값 제거

$$X(t) = A * S(t) \quad \begin{array}{l} \text{측정된 뇌파} \\ \text{신호원 } S(t), \text{ 혼합 매트릭스 } A \end{array}$$

신호원 구하기



통계적으로 비정규성이 최대가 되는 A^{-1} 값

$$A^{-1} \cdot X(t) \cong W \cdot X(t) = S'(t)$$

Database Open Access

EEG Motor Movement/Imagery Dataset

Published: Sept. 9, 2009. Version: 1.0.0

EEG Motor Movement/Imagery Dataset (Sept. 9, 2009, midnight)

A set of 64-channel EEGs from subjects who performed a series of motor/imagery tasks has been contributed to PhysioNet by the developers of the BCI2000 instrumentation system for brain-computer interface research.

When using this resource, please cite the original publication:

Schalk, G., McFarland, D.J., Hinterberger, T., Birbaumer, N., Wolpaw, J.R. BCI2000: A General-Purpose Brain-Computer Interface (BCI) System. IEEE Transactions on Biomedical Engineering 51(6):1034-1043, 2004.

Please include the standard citation for PhysioNet: (show more options)

Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation [Online]. 101 (23), pp. e215-e220. RRID:SCR_007345.

Share

Share icons: Email, Facebook, Twitter, YouTube, LinkedIn

Access

Access Policy:
Anyone can access the files, as long as they conform to the terms of the specified license.

Abstract

This data set consists of over 1500 one- and two-minute EEG recordings, obtained from 109

Files

Total uncompressed size: 3.4 GB.

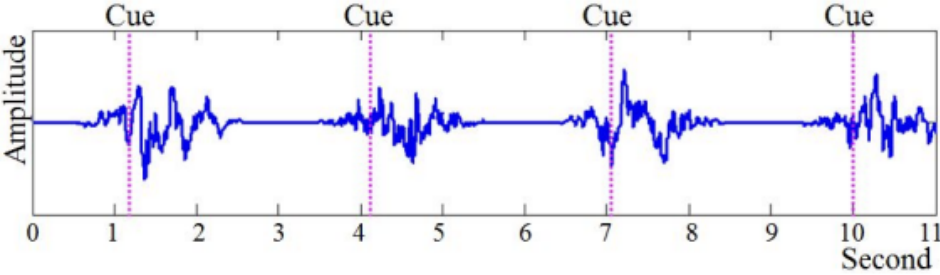
Access the files

- Download the ZIP file (1.9 GB)
- Download the files using your terminal: `wget -r -N -c -np https://physionet.org/files/eegmddb/1.0.0/`
- Download the files using AWS command line tools: `aws s3 sync --no-sign-request s3://physionet-open/eegmddb/1.0.0/ DESTINATION`

Visualize waveforms

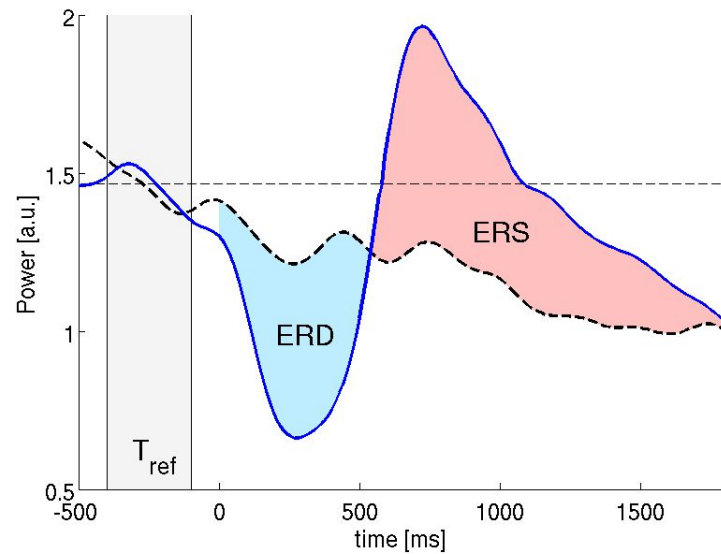
Folder Navigation: <base>	
Name	Size
S001	
S002	
S003	
S004	
S005	
S006	
S007	
S008	
S009	
S010	
S011	
S012	

```
for r_code, label in [("R07", 0), ("R08", 1)]: |
    edf_file = f"{subj_folder}{r_code}.edf"
    edf_path = os.path.join(subj_path, edf_file)
```

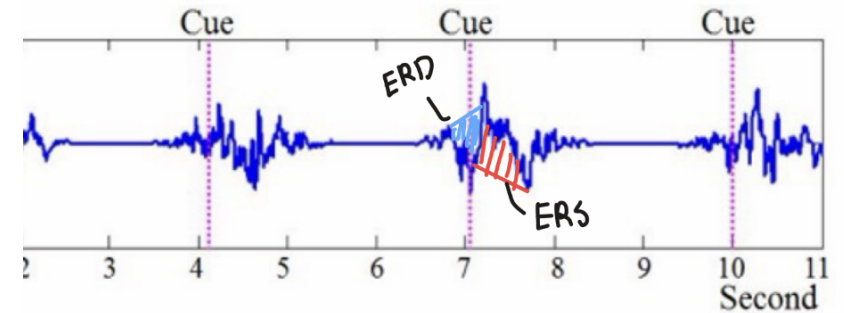


오른쪽 상상 왼쪽 상상 반복

ERD / ERS

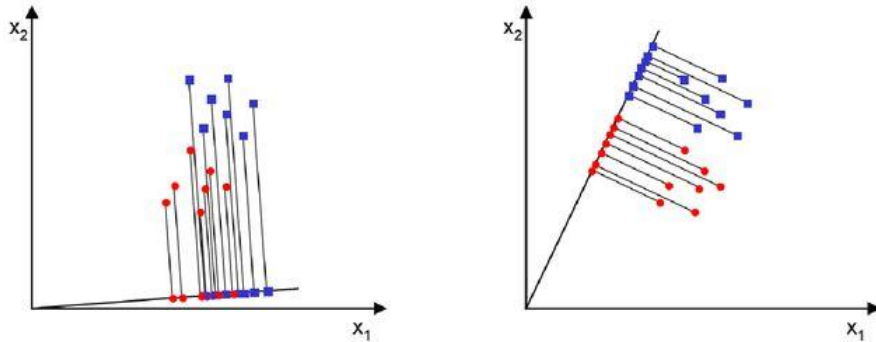


ERD : 주파수 대역이 활성화되며 리듬이 깨지는 현상
ERS : 에너지가 증가하거나 진폭이 커지는 현상



AI 모델

LDA (선형 판별 분석)

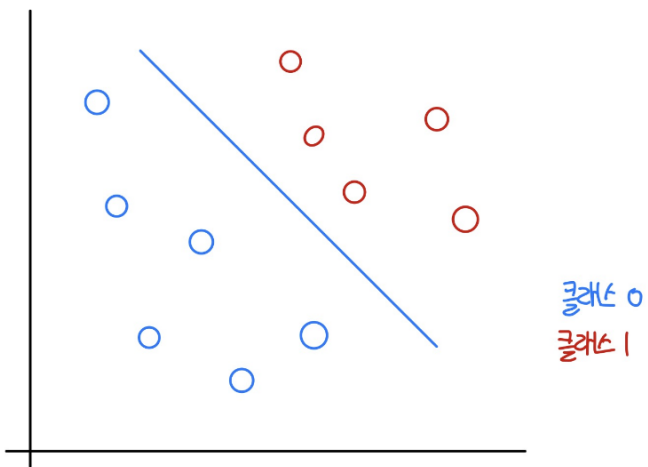


오른쪽 분류가 잘 되어있음

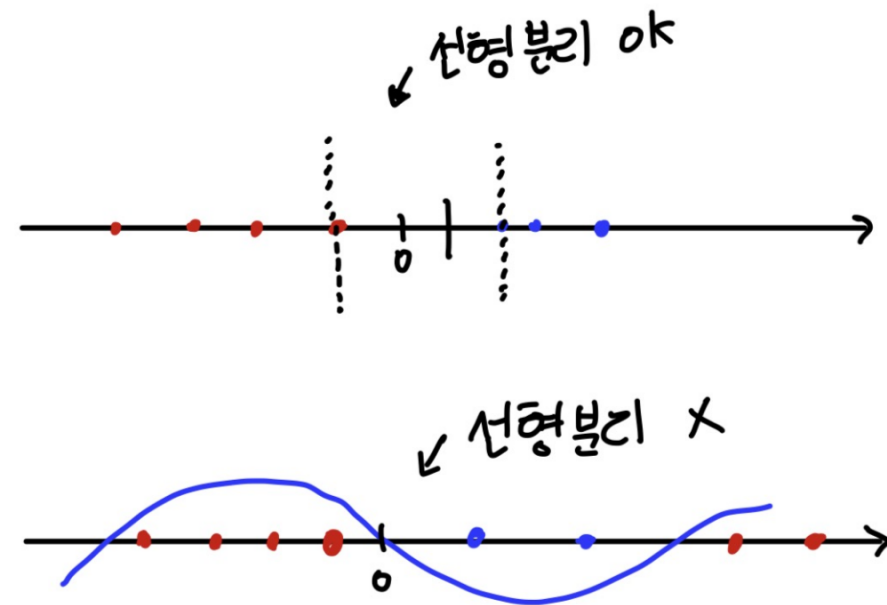
-> 정확도 & 텍스트 위주 모델이기 때문에 한계 존재

투영 후 두 클래스 간 분산은 최대한 크게 가져가고, 클래스 내부의 분산은 최대한 작게 가져가는 방식

SVM 모델



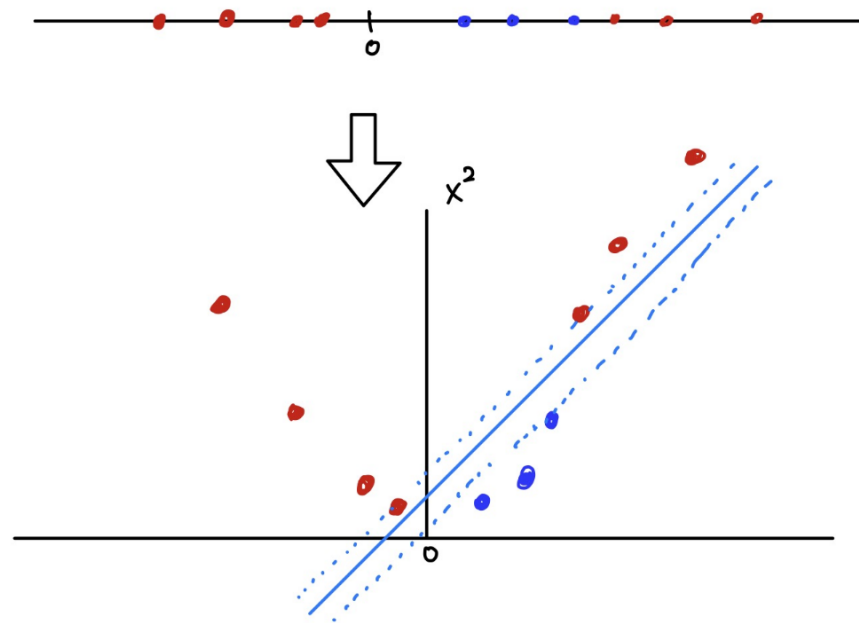
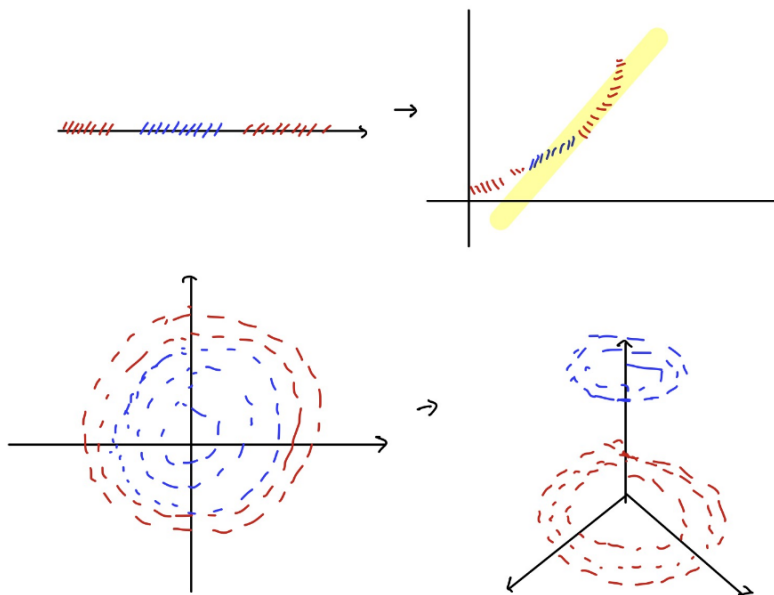
BUT 선형분리 아닐 때

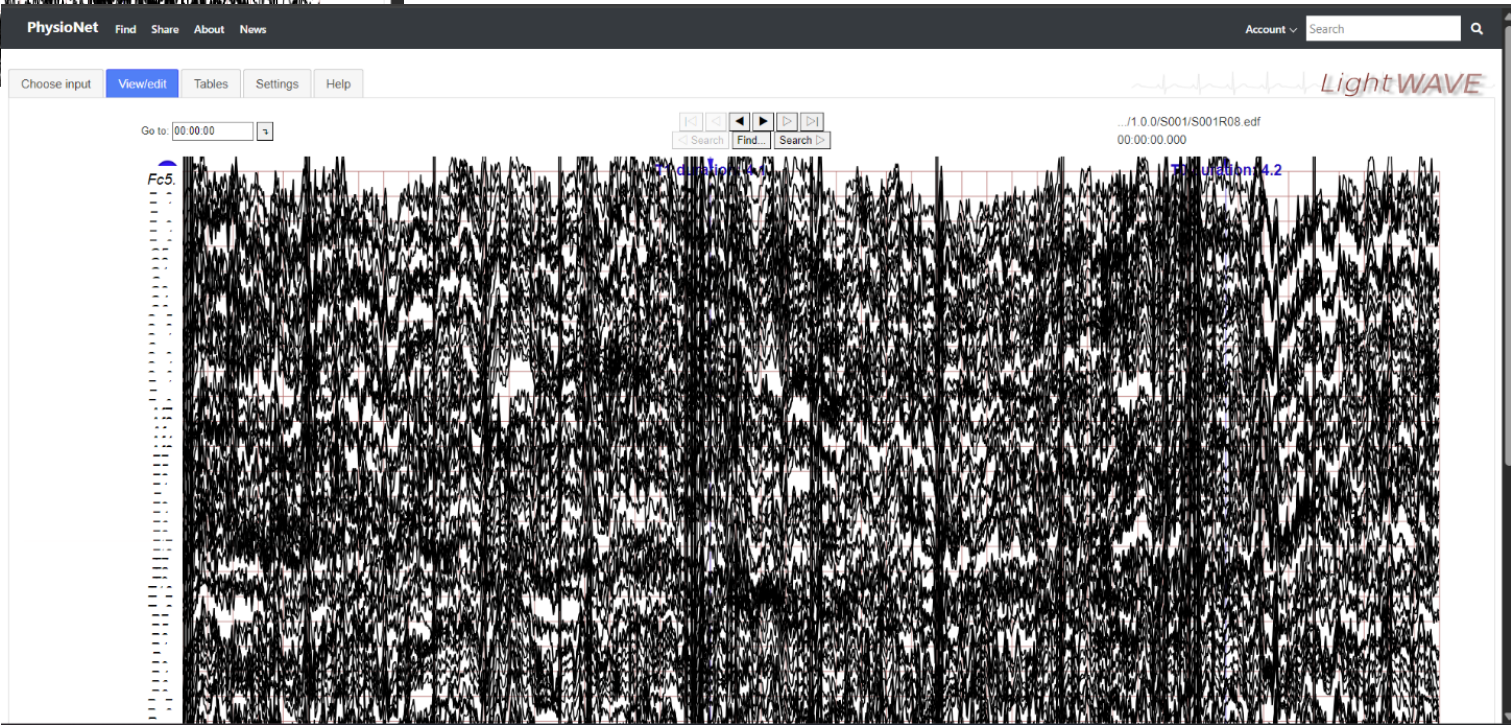
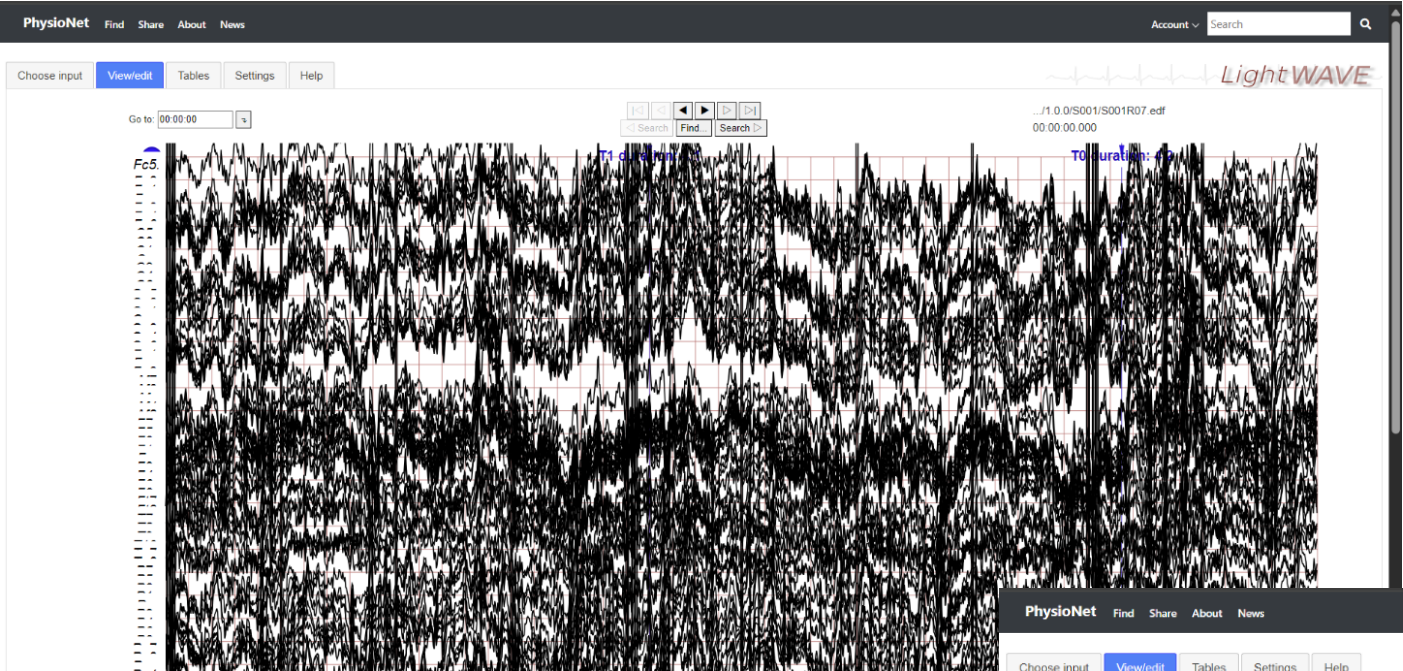


비선형 SVM Model (RBF Kernel SVM)

$$K(x_i, x_j) = \exp[-\gamma \|x_i - x_j\|^2]$$

커널을 이용해 고차원으로 매핑





	A	B	C	D
1	C3...	Cz...	C4...	label
2	7.01E-06	1.48E-07	2.72E-06	0
3	-1.98E-07	3.79E-06	1.55E-06	0
4	1.19E-05	1.20E-05	1.24E-05	0
5	-1.16E-05	-8.50E-06	-1.24E-05	0
6	1.03E-05	5.54E-06	8.61E-06	0
7	-1.90E-05	-1.87E-05	-1.61E-05	0
8	1.12E-05	1.67E-05	1.18E-05	0
9	-4.13E-06	-1.03E-05	-5.36E-06	0
10	1.04E-05	1.15E-05	7.50E-06	0
11	-6.77E-06	-8.72E-06	-8.28E-06	0
12	7.35E-06	1.34E-05	8.62E-06	0
13	-3.41E-06	-7.02E-06	-1.21E-06	0
14	-1.82E-07	3.06E-06	7.00E-07	0
15	-6.38E-06	-6.59E-06	-8.27E-06	0
16	3.33E-06	1.86E-06	2.19E-06	0
17	-5.74E-06	-6.14E-06	-5.36E-06	0
18	4.02E-06	1.19E-06	9.52E-07	0
19	5.24E-06	9.23E-06	8.19E-06	0
20	-6.45E-06	-6.59E-06	-1.57E-06	0
21	-3.61E-07	-5.31E-07	-2.61E-06	0
22	-6.70E-06	-7.72E-06	-1.29E-06	0
23	5.74E-06	1.97E-06	-4.34E-06	0
24	1.57E-05	2.03E-05	1.91E-05	0
25	-1.41E-05	-1.86E-05	-2.02E-05	0
26	1.14E-05	8.99E-06	1.05E-05	0
27	-6.04E-06	-1.99E-06	1.97E-06	0
28	-1.96E-07	2.39E-06	-1.19E-06	0
29	2.43E-06	-2.89E-06	-8.23E-07	0
30	-4.61E-06	-3.99E-06	-2.34E-06	0
31	-1.54E-06	-6.84E-07	-7.39E-07	0
32	4.08E-06	1.67E-06	1.09E-06	0
33	-1.40E-05	-4.90E-06	-3.37E-06	0
34	8.39E-06	-1.21E-06	1.57E-06	0

eeg_features.csv

```

Not setting metadata
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s finished
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s finished
123 matching events found
No baseline correction applied
0 projection items activated
Using data from preloaded Raw for 123 events and 160 original time points ...
0 bad epochs dropped
Processing: /content/drive/MyDrive/files/S109/S109R07.edf
Extracting EDF parameters from /content/drive/MyDrive/files/S109/S109R07.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Reading 0 ... 19679 =   0.000 ... 122.994 secs...

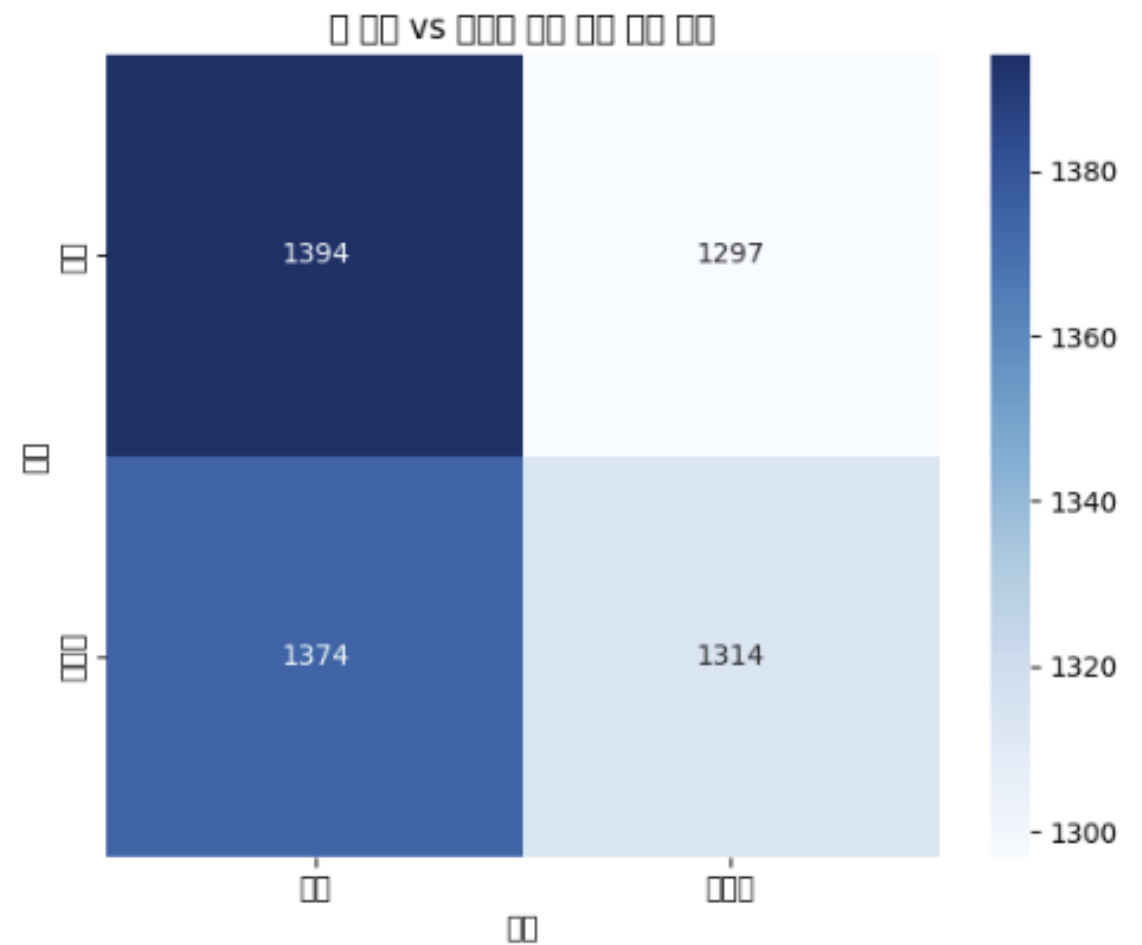
```

```
import joblib
import numpy as np

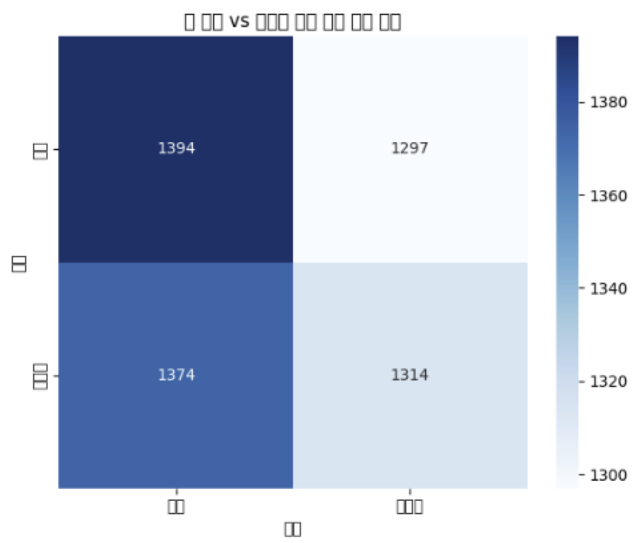
model = joblib.load("/content/eeg_svm_model.pkl")
new_data = np.array([
    [0.1, 0.4, 0.8],
    [0.5, 0.2, 0.1]
])

predictions = model.predict(new_data)
for i, pred in enumerate(predictions):
    label = "왼손" if pred == 0 else "오른손"
    print(f"{i+1}: 예측 결과 → {label}")
```

1: 예측 결과 → 왼손
2: 예측 결과 → 왼손



정확도

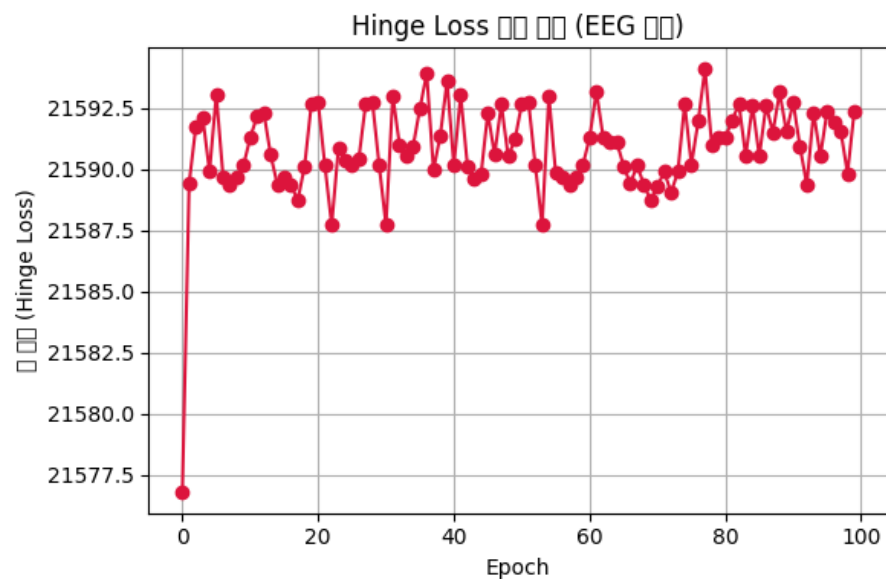


TP/TN/FP/FN

$$\frac{1394+1314}{1394+1297+1374+1313}$$

대략 0.612정도의 정확도

LOSS 계산



경사하강법을 사용하여 손실 함수 그래프

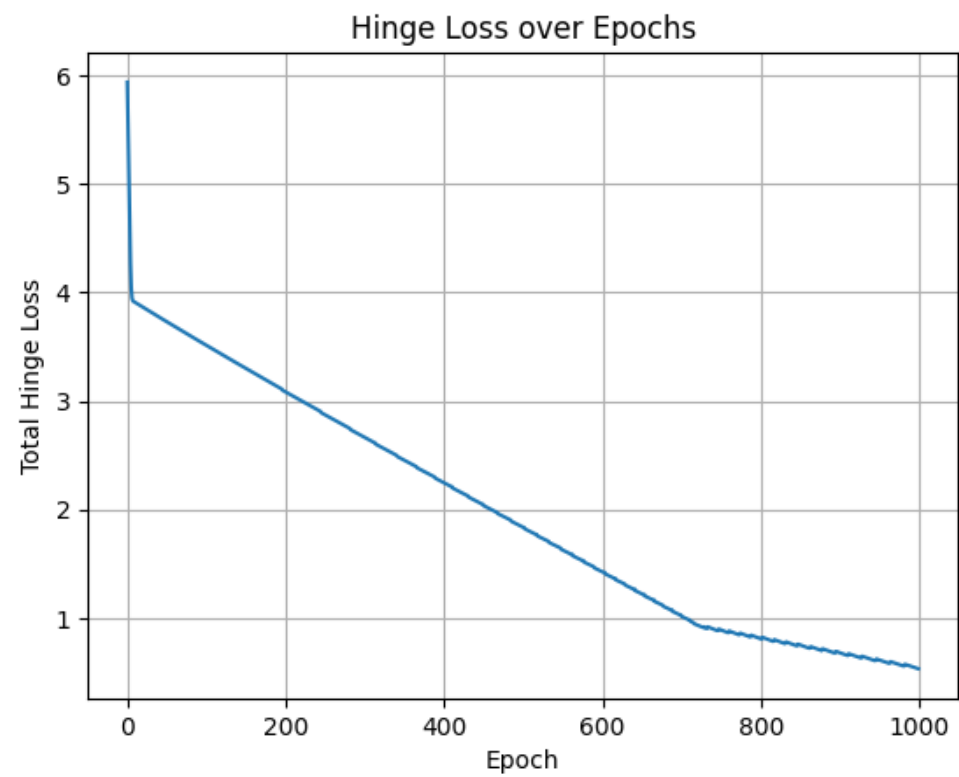
$$\frac{\partial C}{\partial w} = \begin{cases} 0 & \text{if } y \cdot f(x) > 1 \\ -y \cdot x & \text{if } y \cdot f(x) < 1 \\ \text{any value in } [-y \cdot x, 0] & \text{if } y \cdot f(x) = 1 \end{cases}$$

$$\nabla_w = \begin{cases} w, & \text{if } y_i (w^T x_i + b) \geq 1 \\ w - C y_i x_i, & \text{if } y_i (w^T x_i + b) < 1 \end{cases}$$

(w에 대한 서브그래디언트)

$$\nabla_b = \begin{cases} 0, & \text{if } y_i (w^T x_i + b) \geq 1 \\ -C y_i, & \text{if } y_i (w^T x_i + b) < 1 \end{cases}$$

(b에 대한 서브그래디언트)



예측 결과: [-1. 1.]

점점 0으로 수렴한다

HW + AI

기대효과



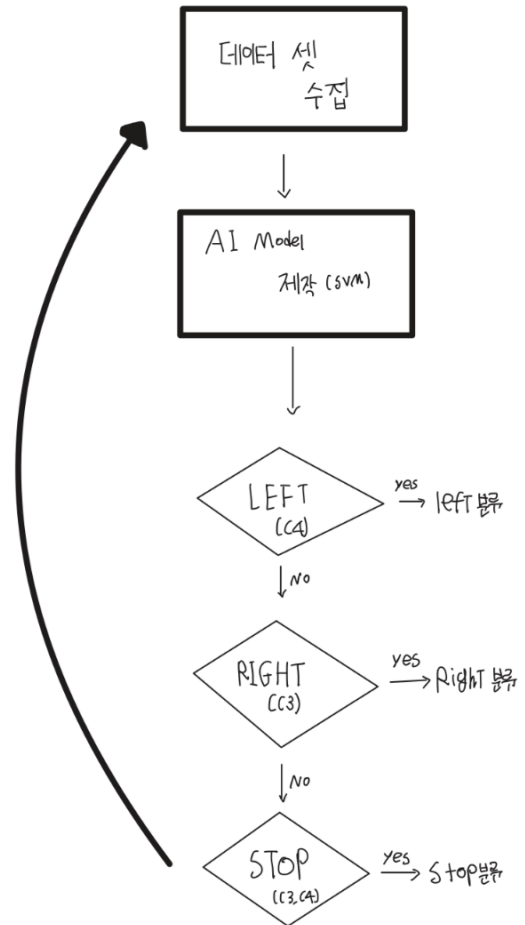
한계점

AI 모델의 정확도

	예측: Negative	예측: Positive
실제: Negative	TN (True Negative)	FP (False Positive)
실제: Positive	FN (False Negative)	TP (True Positive)

뇌파 데이터 + @

결론



감사합니다